Neuroscience & Me Portfolio

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Deliverable2

Introduction to Depression

Depression is a mood disorder that causes a persistent feeling of sadness and loss of interest in things and activities you once enjoyed (Cleveland Clinic, 2023.). Before I learned more about depression, I didn't take it as seriously as I thought it would, such as feeling bad, losing motivation. So, I thought that if I just stayed positive and had a good time, like spending time with friends, I would feel better soon. However, there are many different types of depression, such as clinical depression (the main one), premenstrual dysphoric disorder, and seasonal depression (Cleveland Clinic, 2023), and seeing so many people going through such difficult times made me want to explore depression further. On this page, we will explore depression from a neuroscientific perspective and see how it actually affects.

Neurotransmitters & Depression

First, neurotransmitters can be described as the brain's chemical messengers. Nerve cells in the brain use neurotransmitters to communicate with each other (Schimelpfening, 2024). Our brains communicate between neurons through chemicals called neurotransmitters, which play a key role in regulating mood. Many different neurotransmitters have been identified, of which serotonin is the most widely studied neurotransmitter in depression (Hasler, 2010). It also regulates sleep, appetite, and mood, and plays a role in reducing pain. Studies have found that some people with depression have reduced serotonin signaling, and low levels of this by-product are associated with an increased risk of suicide (Harvard Health Publishing, 2022).

Additionally, neurotransmitters such as dopamine and norepinephrine are also closely linked to depression. Dopamine creates the positive feelings associated with reward and reinforcement, influencing our motivation to continue a task or activity, while norepinephrine acts like adrenaline in the fight-or-flight response and helps send messages to the next nerve cell (Schimelpfening, 2024).

- Fight-or-Flight Response: Known as the acute stress response, refers to the physiological reaction that occurs when in the presence of something mentally or physically terrifying. (Cherry, 2024)

An imbalance of these neurotransmitters can cause or exacerbate symptoms of depression.

Brain regions affected by Depression

One of the brain regions linked to depression is the hippocampus, which controls memory and emotion regulation. Studies have shown that some depressed people have smaller hippocampus. For example, one FMRI study showed that, on average, the hippocampus was 9% to 13% smaller than women who were not depressed. Experts also say that constant exposure to stress hormones impairs the growth of neurons in the hippocampus part of the brain (Harvard Health Publishing, 2022).

Of course, in addition to the hippocampus, the effects of depression on the brain are multifaceted and include the prefrontal cortex, thalamus, and caudate nucleus. In some brain images of depressed people, they have found abnormally low prefrontal cortex, and the thalamus, the part of the brain that receives the most sensory information, can be affected by depression by associating it with negative emotions (Rosenblatt, 2021).

Conclusion & key takeaways

Knowing these facts helped me understand that depression is not just about the emotions. It is also biological and involves real changes in the brain. Before, I thought depression was just about feeling sad or not having energy. But I learned that an imbalance in brain chemicals like serotonin, dopamine, and norepinephrine can strongly affect how we feel and behave.

I also learned that parts of the brain like the hippocampus and the frontal lobe are closely connected to depression. Stress and other things around us also have a bigger effect than I thought. This helped me realize that depression is not just about someone being weak or lazy.

Deliverable 3

Introduction to neuroscience in marketing

My major is Digital Marketing, and I started get more interested in how psychology and neuroscience affect advertising. I don't have specific plan for future yet, but I like to learn about marketing strategy, especially about branding and social media. So I became curious how people make decisions when they see marketing, and how neuroscience explain that process. After learning some in class, I found something called neuromarketing, and it was very interesting for me. This interest led me to explore neuromarketing, which combines neuroscience, psychology, and marketing.

Role of neuromarketing

Neuromarketing uses neuroscience tools such as EEG, eye tracking, and FMRI to observe brain activity and physiological responses in real time. This allows us to learn hidden preferences through the brain activity of consumers who may not be able to clearly tell if they prefer one brand over another. For example, neuromarketing can tell us which elements of an advertisement most appealing, which elements are trigger emotional responses, and how product design influences them. (Bridgewater, 2024)

This approach provides data beyond traditional surveys or focus groups. Instead of asking consumers what they think, neuromarketing measures how they actually respond, providing deeper insight into their preferences and behaviors.

Emotional & memory impact in advertising

According to Harvard Business Review, consumer emotions play a key role in purchasing decisions. The amygdala of the brain functions to process emotions, and ads that induce a strong emotional response tend to be remembered longer for the brand (Harrell, 2019). Also, repeated exposure is essential to imprint a brand in consumers' memories. According to the American Marketing Association (AMA), the more frequently an advertisement is shown, the more likely it is that the brand will be stored in consumers' long-term memory (Hoffman Marketing Communications, 2014)

In particular, the effect is greater when visual and auditory elements such as colors, logos, advertising music, and slogans are consistent. Promotions such as limited-edition product releases, discounts, coupons, and free shipping on first purchases are elements that stimulate consumers' desire to purchase. These marketing techniques can stimulate the consumer's dopamine secretion and induce impulse purchases (Harrell, 2019).

Neuroscience knowledge has contributed to several advancements in digital marketing. Considering that repeated exposure to a brand can increase the likelihood that will be stored in long-term memory by the consumer's hippocampus, I will take this into consideration and keep their brand colors and logo consistent to make it easily recognizable to consumers, and will provide repeated exposure not only through advertising but also through social media, YouTube, etc. It was also interesting to learn that analyzing where consumers' eyes are focused on an ad can help optimize ad design.

Practical applications of digital marketing

As these factors clearly have a significant impact on brand growth, that optimize ad placement to improve readability by placing it where consumers' eyes are likely to go, such as CTA (Call to Action) buttons, brand logos, and core content. Based on the materials I have referred to; they all emphasize that neuromarketing technology is driving the development of digital marketing and that its use can maximize advertising effectiveness by analyzing consumers' unconscious responses.

Conclusion & future implications

I think that the major advantage of developing such technology is that it will be able to provide immediate feedback on advertisements. It was also very interesting to be able to measure consumers' EEG responses while they were viewing an ad and analyze which elements elicited a strong response (Hoffman Marketing Communications, 2014). While this has the advantage of allowing for more objective advertising optimization, it can also have drawbacks such as the high cost and the difficulty of configuring the experimental environment. I found it interesting that digital marketing is not just about creativity and trends but is connected to the way the human brain processes information and emotions.

Neuroscience of wellness

This quarter, I decided to focus on improving my sleep with wellness techniques. I don't have a good sleep pattern to begin with, but since coming to the US, it has become more irregular, and I have felt tired more often. First, before trying these patterns, if you look at my original lifestyle pattern, I will go to sleep at 2-3am and wake up at 8:30 if I had school and wake up past 10 if I didn't. Also, after eating around 3, I often had to take a nap for about two hours. To improve these sleep patterns, I tried the following pattern:keep the bedroom cool, avoid naps, and create a bedtime routine. I tried this pattern for a week starting the first week of March.

As always, the thing that helped me the most was to avoid napping. In particular, I tried to overcome the post-meal sleepiness by forcing myself to do something to move my body and wake up from the sleepiness. I also made sure to prepare myself for bed by getting ready in advance. Also, by preparing in advance, I had time to prepare for bed. This improved the quality of my sleep, and I often woke up naturally in the morning without an alarm. Also, seeing that a cool environment helps people fall asleep, which was the most interesting finding in the study, I didn't turn up the temperature before falling asleep. When I slept with the temperature turned up, I often woke up feeling too dry and had a hard time, but that has improved a lot. I'm also trying to be more mindful of stress management, but it still seems difficult. This change has helped my overall sleep quality. (Manning, 2021)

Neuroscience Analysis

Sleep is really connected to how our brain works. One important part is called the suprachiasmatic nucleus (SCN), and it controls our body clock, which is also called circadian rhythm (Nordqvist, 2018). This part helps to release melatonin, the hormone that reacts to light and makes us sleepy. I realized that I could fall asleep easier when my room was dark and cool at night because of this reason. Also, the frontal lobe, which helps us control emotions and focus, is affected when we don't sleep enough. If we lose sleep, we can feel more stressed, hard to concentrate, and have memory problems. These brain parts don't work well without sleep, and then our body feels out of balance.

Something I found interesting was that even if we feel sleepy in the afternoon, like between 1 and 3pm, taking a nap doesn't really fix bad sleep at night. (Nordqvist, 2018) It's best to try sleeping 7–8 hours at night, even if it's hard. If we nap for like over 30 minutes, it can make it harder to sleep at night again.

Why do we eat?

"Why do we eat?" This question is really simple, but it is one that causes hesitation and deep thought the moment you hear it. Almerico argues that unlike animals, humans have a complex and unique way of eating, including foraging, hunting, and using cooking utensils, and that humans do not simply ingest, but eat. As Almerico argues, humans do not simply consume, but also perform the action of eating. We do not only eat, but also have complex cultures such as cooking, sharing, and observing etiquette (Almerico, 2014).

For example, in Korea, there is a word for "식구(shik-gu).” This word means “people who eat together,” and it reflects the Korean cultural that sharing a meal together is not simply a blood relationship, but also a family. Almerico’s argument aligns well with this concept, as it highlights that eating is not just about survival but also about forming relationships. In this way, dining goes beyond simple sustenance; it carries social significance and serves as an essential part of human connection.

In this way, we don’t just simply eat to solve hunger, but are influenced by various factors including emotions, the environment, and culture. Unlike animals, humans form culture through food. And as reflected in the common Korean farewell we say, “Let’s have a meal together next time,” and we see once again that we form culture and create social relationships through food.

Why can’t we sleep?

Lack of sleep can literally steal years from our lives and is directly linked to obesity, diabetes, increased cancer risk, and neurodegenerative diseases like Alzheimer's. Plus, insomnia often leads to depression, and depression can lead to insomnia: UCSF sleep expert Aric Prather, PhD, has shown conclusively that the longer people stay asleep, the more likely they are to get sick. (Manning, 2021)

The most surprising result of the chronotype experiment was that the later the midpoint of sleep, the higher the rate of depression, smoking, and drinking. When they think about "why are we experiencing such suffering?" despite these effects, we find that it is an effect that focuses on morning productivity. This make me feel how big the social impact is. (Harvard Health Publishing, 2021)

The current structure of work and school is mostly set up in favor of morning people. Most start at 8-9am, because people think they are most productive in the morning. However, this does not apply to everyone, and it is a very difficult environment, especially for night people. If night people follow the morning pattern, they may be less efficient because they have to work or study with their brain not fully awake. I think our sleep needs and work/school needs don't match up, people whose needs fit seem rare, and I think most people live with reducing their own sleep to accommodate work/school needs.

Also, one surprising thing is that the World Health Organization has even designated shift work as a cause of cancer. This also highlights the big impact that sleep has on us. (Harvard Health Publishing, 2021)

Self-assessment & reflection

Working on my portfolio this quarter has given me the opportunity to see how neuroscience applies to everyday life. Through each component of Neuroscience Foundations Midterm, Neuroscience and My Career, and Wellness, I was surprised to see how the brain is not only connected to mental health, but also to everyday behaviors, emotions, and even marketing strategies.

For my midterm project, I explored depression to discover how neurotransmitters such as serotonin, dopamine, and norepinephrine affect brain regions such as the hippocampus and frontal lobe in people with depression, helping me understand that depression is not just emotional, but biological and structural.

For neuroscience and my career, I connected with my major, digital marketing. It was interesting to see how neuromarketing uses brain research to make advertising more effective. For example, I learned how the amygdala and hippocampus respond and how they help store memories, which made me realize that marketing is not just creative, but scientific. This fact was so new to me and made me want to explore more.

Finally, the wellness component gave me a first-hand experience. I observed my sleep patterns and tried to implement strategies to improve my sleep quality, such as not taking naps and maintaining a good bedroom temperature at night. This improved my sleep quality and was an important experience for me. I also learned about the fact that the superior cross phasic nucleus (SCN) controls the sleep cycle and influences emotion regulation, memory, and stress.

What was most interesting about this portfolio was that it connected neuroscience with my major, digital marketing. I thought advertising and brand strategy was just focused on design and phrases, but it was fun to see what happens in the brain when people see an ad, and how emotions and memories affect consumer behavior. I also realized that neuromarketing is a scientifically designed strategy in that it can steer consumer behavior through repeated exposure and emotional messages.

After learning about these brain reactions, I realized that in the future, when I work with advertising or in marketing, I want to create marketing that is not just "pretty and eye-catching," but that will leave a lasting impression in people's hearts, empathize more deeply with consumers, and draw out what they really want. Since marketing is ultimately about understanding people's emotions and behavior, I felt that it was very important to understand the underlying mechanisms of the brain and the psychological principles that underpin it. I also heard the idea that if we use these learnings in our actual marketing strategies, I can communicate more authentically. Rather than simply focusing on the number of clicks or views, I can have a better impact by interpreting and reflecting why people responded to the ad and what they felt.

The most difficult part of researching these materials and obtaining and learning new information was organizing, understanding, and creating all of this in English. This was especially challenging because this was my second psychology class and also second quarter, and I was still unfamiliar with many of the vocabulary words. It was also difficult to keep track of all the research sources and citations. However, it was interesting to see real-life cases, and since English is not my native language, I was able to deal with perspectives from various countries.

If I could do it over again, I would try to find a little more material because the more material I found, the more facts I knew and the more interesting things I could see. I would also have maintained my sleep pattern for a little longer. I guess it's more memorable for me because I experienced it firsthand. Of course, I know I can start again anytime, so I will try to practice well during this break and try to be in good shape to listen to the classes in the upcoming spring quarter.

One skill I developed in this class was taking online courses. This was my first online class, so it was more challenging than I thought. I think the two main reasons were finding my own time to attend the course and staying focused. To do this, I planned to choose a time when I was most focused and alert and attend the course consistently. For me, that time was 8 p.m. Most people might think that it's too late, but it was a good time for me because my sleep pattern wasn't really established yet. I currently live in a dorm, and I've found that I can concentrate better when I go out to the lounge and listen to classes, so I can create own routine. Also, I listened to the lectures at a slower speed than the regular speed, and when I had to do assignment, I played the lecture again and did the homework. After listening to the lectures this way, if there is something I would recommend, instead of listening to the lectures at double speed because you want to listen to them quickly, you will find that if you listen to the lectures at a slower speed, your understanding will increase, and since you will not have to go back and watch the lectures again.

I think this portfolio has value beyond the classroom. I know that psychology is a required course in every department, and I think it is worth it. It has already really affected my life as I started to adjust my sleeping habits and most importantly, it has affected my thinking about my future. It has made me think about how I can implement my own marketing beyond just working in a marketing company and I will be more conscious of how psychology can affect my future career.

I also want to share the wellness-related information I learned with my friends and family. The biggest thing I gained from this class was that I could explain things based on brain science rather than just giving advice.

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